



Department of Mathematics and Natural Sciences
MAT110 : Differential Equations and Coordinate Geometry
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Successive Differentiation

1. If $x = \sin\sqrt{y}$, then show that, $(1 - x^2)y_2 - xy_1 = 2$.
2. If $y = e^{\tan^{-1}x}$, then show that, $(1 + x^2)y_2 + (2x - 1)y_1 = 0$.
3. If $\ln y = \tan^{-1}x$, then show that, $(1 + x^2)y_2 + (2x - 1)y_1 = 0$.
4. If $\ln y = a \sin^{-1}x$, then show that, $(1 - x^2)y_2 - xy_1 - a^2y = 0$.
5. If $y = \tan^{-1}x$, then show that, $(1 + x^2)y_2 + 2xy_1 = 0$.
6. If $y = \cot^{-1}x$, then show that, $(1 + x^2)y_2 + 2xy_1 = 0$.
7. If $y\sqrt{1 - x^2} = \sin^{-1}x$, then show that, $(1 - x^2)y_2 - 3xy_1 - y = 0$.
8. If $y = e^{m \sin^{-1}x}$, then show that, $(1 - x^2)y_2 - xy_1 - m^2y = 0$.
9. If $y = e^{m \cos^{-1}x}$, then show that, $(1 - x^2)y_2 - xy_1 - m^2y = 0$.
10. If $\tan(\ln y) = x$, then show that, $(1 + x^2)y_2 + (2x - 1)y_1 = 0$.
11. If $\ln y = m \cos^{-1}x$, then show that, $(1 - x^2)y_2 - xy_1 - m^2y = 0$.